

Maximal Lactate Accumulation Rate in All-out Exercise Differs between Cycling and Running

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ABSTRACT

This study aims to compare maximal lactate accumulation rate ($\dot{V}La_{max}$) and power output (P_{max}) between cycling and running in terms of reliability, differences between, and correlations among modalities. Eighteen competitive triathletes performed a 15-s all-out exercise test in cycling and a 100-m sprint test in running. Each test was performed twice and separated by one week. Exercise tests in cycling were performed on an ergometer whereas sprint tests in running were performed on an indoor track. Differences between trials and exercise modality were analyzed using two-way ANOVA. $\dot{V}La_{max}$ (ICC = 0.894, ICC = 0.868) and P_{max} (ICC = 0.907, ICC = 0.965) attained ‘good’ to ‘excellent’ reliability in both cycling and running, respectively. $\dot{V}La_{max}$ was higher in running ($d = 0.709$, $p = 0.016$) whereas P_{max} was lower in running ($d = -0.862$, $p < 0.001$). For $\dot{V}La_{max}$, limits of agreement between modalities ranged from -0.224 to $+0.437$ $mmol \cdot l^{-1} \cdot s^{-1}$. P_{max} correlated between modalities ($r = 0.811$, $p < 0.001$), whereas no correlation was found in $\dot{V}La_{max}$ ($r = 0.418$, $p = 0.084$). $\dot{V}La_{max}$ is highly reliable in both modalities and higher in running compared to cycling. Since $\dot{V}La_{max}$ does not correlate between cycling and running, it should be determined sport-specifically.

Introduction

Endurance exercise performance depends on the athletes’ metabolic profile that can be described by several parameters: maximal oxygen consumption ($\dot{V}O_{2max}$), the percentage of $\dot{V}O_{2max}$ at lactate threshold and movement efficiency [1, 2]. These aerobic parameters can be derived from incremental step tests with different rates of increments by using gas exchange analyses [3]. In addition to these major determinants, maximal lactate accumulation rate ($\dot{V}La_{max}$) was found to be a promising parameter to describe the power of the glycolytic metabolism, which is important for starts, accelerations and the sprint finish of a race [4–7]. $\dot{V}La_{max}$ is derived from sprint exercises lasting approximately 10 to 15 s [4]. Mathematical simulation approaches assume that the oxidative and gly-

colytic metabolism (in terms of $\dot{V}O_{2max}$ and $\dot{V}La_{max}$, respectively) interact to determine the equilibrium of lactate production and removal, which is usually referred to as the maximal lactate steady-state [8, 9]. Maximal lactate steady-state was found to be one of the most valid lactate threshold parameters, as it is highly correlated with endurance performance [10]. However, the gold-standard method of determining maximal lactate steady-state (by using several constant load trials of 30 min) is rather time consuming, restricts the training schedule of endurance athletes, and does not explain why maximal lactate steady-state occurs at a certain intensity (e. g. expressed as a percentage of $\dot{V}O_{2max}$) [11, 12]. Hence, approaches focusing on the distinction of oxidative and glycolytic power seem to be promising for providing a more detailed meta-

bolic profile of the athlete, which might be helpful for predicting endurance performance on a physiological basis. In fact, recent research demonstrated that training-induced improvements in the calculated maximal lactate steady-state arise from different alterations in $\dot{V}O_2\text{max}$ and $\dot{V}La_{\text{max}}$ [13].

Multisport athletes – such as triathletes or duathletes – who undergo simultaneous training in cycling and running may demonstrate differences in their oxidative and glycolytic metabolism between these exercise modalities [14]. For $\dot{V}O_2\text{max}$, triathletes demonstrated slightly higher [15] or rather similar [16] values in running compared to cycling exercise. However, differences and adaptations in $\dot{V}O_2\text{max}$ seem to depend on the athletes' training background in the particular modality [17, 18]. Parameters associated with the glycolytic metabolism in terms of post-exercise lactate concentration and maximal accumulated oxygen deficit demonstrated inconsistent differences between cycling and running [19–21]. Peak power output (P_{max}) attained in the Wingate anaerobic test on a cycle ergometer was not associated with sprint performance in running, indicating the need for sport-specific testing [22].

$\dot{V}La_{\text{max}}$ has already been determined by using all-out exercise tests in cycling [5, 9, 13, 23], handcycling [6, 23], running [7, 24] and canoe polo [25]. However, the effect of exercise modality on $\dot{V}La_{\text{max}}$ has purely been examined in a within-participant design. Just recently, $\dot{V}La_{\text{max}}$ and P_{max} were found to be extremity-specific, since both parameters were higher in cycling when compared to handcycling and demonstrated no significant correlation between extremities [23]. This indicates that examining the specificity of $\dot{V}La_{\text{max}}$ allows for more profound insights into physiological differences of the glycolytic metabolism and establishing modality-specific metabolic profiles in multisport athletes. Recently, a sport-specific field test to determine $\dot{V}La_{\text{max}}$ and P_{max} in running has been developed, which demonstrated high reliability and association with sprint performance [7]. Hence, the aims of this study were to compare $\dot{V}La_{\text{max}}$ and P_{max} between cycling and running in terms of reliability, differences between and correlations among exercise modalities.

Materials and Methods

Participants

This study is part of a larger project focusing on the specificity of $\dot{V}La_{\text{max}}$. As mentioned earlier, parts of this project have already been published elsewhere, which describe the differences between upper and lower extremity by using exercise tests in handcycling and cycling, respectively [23]. This study focusses on the effect of exercise modality by comparing exercise tests in cycling and running. The participants described below are the same for both studies.

Eighteen ($n = 3$ female, $n = 15$ male) competitive triathletes (25.1 ± 2.8 yrs., 1.79 ± 0.08 m, 71.6 ± 6.3 kg) participated in this study. Body fat was determined using ten-site skinfold thickness measures. The mean body fat percentage of the participants was 14.4 ± 3.7 %. The participants were used to a weekly training routine of 11.6 ± 4.3 h $\cdot$ wk $^{-1}$ and had participated in triathlon compe-

titions for 6.7 ± 4.7 years. Prior to any testing, the participants were evaluated based on the guidelines of the European Society of Cardiology, which include a review of the family and personal history, a physical examination and a resting electrocardiogram. Afterwards, only the participants without positive findings were included. To ensure standardized conditions between measurements, all participants were instructed to refrain from caffeinated beverages for at least 8 h before testing and avoid vigorous physical activity on the testing days and the day before. To avoid influences based on circadian rhythm, the participants performed the tests at approximately the same time of the day. The study was approved by the local ethics committee (No. 124/2017) and complied with the Helsinki Declaration modified in 1983 and the ethical standards in sport and exercise science research [26]. All participants were informed about the procedures and risks of the study and gave their written consent before participating in this study.

Design

The participants performed two all-out exercise tests in each exercise modality (cycling and running) within a two-week period. In order to familiarize the participants with these tests, each test was performed twice and separated by one week (T1 and T2). To ensure full recovery between tests in cycling and running, exercise tests in cycling were performed on Mondays, whereas exercise tests in running were performed on Fridays. To express power output in a normalized fashion [$\text{W} \cdot \text{kg}^{-1}$], the participants' body mass, was determined immediately before every single exercise test.

In cycling, 15-s all-out tests were performed on the participants' own road bikes that were attached to a fully calibrated and validated ergometer (TE 2 %, Cyclus 2, 8 Hz, RBM electronic automation GmbH, Leipzig, Germany). Prior to the 15-s all-out tests, participants performed a standardized low-intensity warm-up of 10 min (at 100 W), including three acceleration bursts of ten seconds (linearly up to 500 W) [23, 27]. After the warm-up, participants performed a passive rest for 5 min in seated position.

In running, the participants performed a 100-m sprint test on an indoor track, including a standardized warm-up of 15 min as recently published [7]. Throughout the sprints, participants were verbally encouraged by the examiners. The time to cover the 100 meters (t_{100}) was determined using a start pedal and a double infrared photoelectric light barrier [7]. The velocity of the athletes was determined using a laser velocity guard (LDM-300-C-SPORT, 100 Hz, Jenoptik, Jena, Germany), installed as previously published [7].

Instrumentation

The 15-s all-out tests in cycling were performed in isokinetic mode and with cadence limited to 130 min $^{-1}$. The initial resistance of the sprint test in cycling was set to $2.0 \text{ N} \cdot \text{kg}^{-1}$ body weight [23, 28]. Throughout the tests, participants had to remain seated and were verbally encouraged by the examiner. Zero values at the beginning of exercise were erased from raw power data. Raw power data of cycling were interpolated using a modified biexponential time function according to Eq. 1:

$$P(t) = A_p \cdot (1 - e^{-k_1 \cdot t}) - A_p \cdot (1 - e^{-k_2 \cdot t})$$

where $P(t)$ = interpolated power output at a certain time [$W \cdot kg^{-1}$]; t = time [s]; A_p = amplitude parameter describing power output during the 15-s all-out test [$W \cdot kg^{-1}$]; $k_{1,p}$ = velocity constant describing the increase in power output during the 15-s all-out test [s^{-1}] and $k_{2,p}$ = velocity constant describing the decrease in power output during the 15-s all-out test [s^{-1}]. The parameters were determined using a nonlinear regression technique with a criterion of residual convergence lower than 10^{-8} (SPSS, 25, SPSS Inc., Chicago, Ill., USA). The starting values for A_p , $k_{1,p}$ and $k_{2,p}$ were 10, 0.5 and 0.05, respectively.

In running, velocity and power profiles were calculated with regard to the power needed to maintain the height and velocity of the center of mass (P_C), the power needed to change the kinetic energy of the center of mass (P_A) and the power to overcome aerodynamic drag (P_D) according to recent research [7]. A_f was individually determined using calibrated pictures in a running-specific posture (► **Fig. 1**) [7].

For both exercise modalities, blood samples were collected at the arrival of the participants, at the end of the warm-up, immediately before and after the all-out exercise tests, as well as every minute after exercise for 10 min on the right earlobe, using an enzymatic-amperometric sensor chip system (Biosen C-Line, EKF-diagnostic GmbH, Barleben, Germany). Lactate kinetics following all-out exercise tests in cycling and running were interpolated using a modified biexponential time function as described in previous research (**Eq. 2**) [6, 7].

$$La(t) = La_{pre} + A \cdot (1 - e^{-k_1 t}) - A \cdot (1 - e^{-k_2 t})$$

where $La(t)$ = interpolated lactate concentration at a certain time [$mmol \cdot l^{-1}$]; t = time [min]; La_{pre} = resting lactate concentration im-

mediately before the test [$mmol \cdot l^{-1}$]; A = amplitude parameter describing post-exercise lactate kinetics of the all-out test [$mmol \cdot l^{-1}$]; k_1 = velocity constant describing the exchange of lactate from the previously active muscles [min^{-1}] and k_2 = velocity constant describing the removal of lactate during passive recovery [min^{-1}]. The parameters were determined using a nonlinear regression technique with a criterion of residual convergence lower than 10^{-8} (SPSS, 25, SPSS Inc., Chicago, Ill., USA). The starting values for A , k_1 and k_2 were 3, 0.5 and 0.005, respectively.

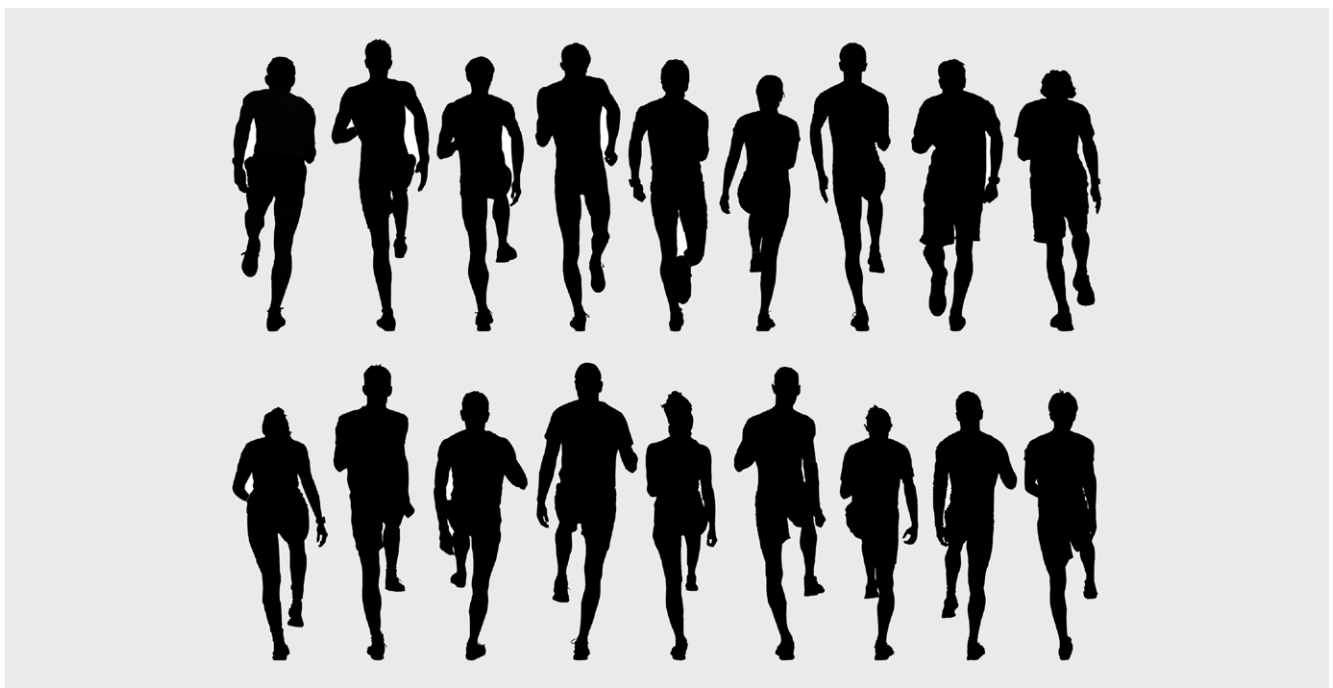
$\dot{V}La_{max}$ was calculated as the difference (La_Δ) between the measured maximal post-exercise lactate concentration (La_{max}) and resting lactate concentration (La_{pre}) that was divided by the difference between test time (t_{test}) and the period at the beginning of exercise for which no lactate formation is assumed (t_{alac}) (**Eq. 3**) [4–7, 9].

$$\dot{V}La_{max} = \frac{La_{max} - La_{pre}}{t_{test} - t_{alac}} = \frac{La_\Delta}{t_{lac}}$$

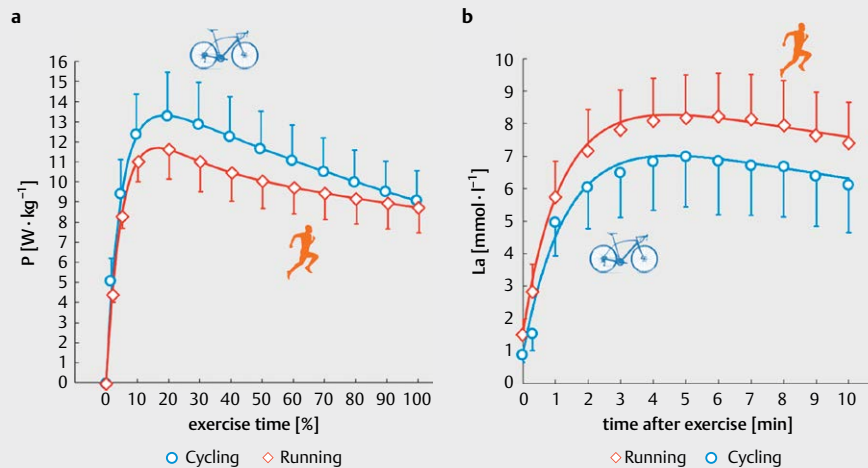
As a representation of phosphocreatine metabolism, t_{alac} was determined as the period when power output decreased by 3.5 % ($t_{pmax-3.5\%}$) [6, 7, 9]. The difference between t_{test} and $t_{pmax-3.5\%}$ can be classified as lactic time interval (t_{lac}).

Statistical analyses

Statistical analyses were done using the Statistical Package for the Social Sciences software (25, SPSS Inc., Chicago, Ill., USA). The reliability of the parameters was analyzed using the intraclass correlation coefficient (ICC, Model: Alpha, two-way random; Type: Absolute agreement, single measures) and the typical (standard) error (TE) [7, 29]. ICCs were classified as ‘excellent’ ($ICC \geq 0.90$), ‘good’ ($0.90 > ICC \geq 0.75$), ‘moderate’ ($0.75 > ICC \geq 0.50$) or ‘poor’



► **Fig. 1** Individual frontal area (A_f) determined using calibrated images of every participant.



► **Fig. 2** Power output during the all-out exercise tests (left) and post-exercise lactate kinetics (right) for cycling and running at T2 **a** Power output [$\text{W} \cdot \text{kg}^{-1}$] with respect to relative exercise time [%] for cycling and running. **b** Post-exercise lactate concentration [$\text{mmol} \cdot \text{l}^{-1}$] with respect to time after exercise [min] for cycling and running. Values are expressed as mean ($\bar{x}$) and standard deviation (SD). La =lactate concentration; P =power output.

(ICC < 0.5). TEs were expressed in absolute values and as a percentage of the mean. To evaluate the parameters in their ability to monitor the athletes' individual development, TEs were compared to the smallest worthwhile change (SWC) [7]. The ratio between TE and SWC (TE/SWC) was calculated with values of < 1.0 indicating an adequate precision. Reliability was examined for body mass, peak power ($P_{\max}$), work accomplished during the all-out exercise tests (W), time to attain peak power ($t_{p\max}$), $t_{p\max-3.5\%}$, t_{lac} , lactate kinetics interpolation parameters (A , k_1 , k_2) and the lactate response (La_{pre} , $La_{\max}$, La_{Δ} , and $\dot{V}La_{\max}$). For $\dot{V}La_{\max}$, differences between trials and modality were quantified using Bland-Altman plots and statistics including the limits of agreement [7, 28].

Differences between trials (T1 and T2) and exercise modality (cycling and running) were examined using a two-way (2×2) analysis of variance (ANOVA). Post-hoc comparisons were based on Bonferroni's correction. To identify differences in t_{100} between trials, a paired t-test was applied. As effect-sizes, partial eta squared (η_p^2) and Cohen's d (d) were calculated. The relationship between $P_{\max}$, W , La_{Δ} and $\dot{V}La_{\max}$ was analyzed using Pearson's correlation coefficient at T2. Beforehand, the Kolmogorov-Smirnov test was applied to check for normality. The level of significance for inferential analyses was set at $\alpha = 0.05$.

Results

Reliability and differences between trials

During the all-out exercise tests, power output demonstrated a rapid increase until about 15–20 percent of total exercise time in both exercise modalities (► **Fig. 2a**). Post-exercise lactate kinetics demonstrated an increase until about 3–4 min after the all-out tests (► **Fig. 2b**). Afterwards, an almost linear decrease in lactate concentration was found in cycling and running.

Body mass, $P_{\max}$ and W attained 'good' to 'excellent' reliability in cycling and running, respectively (► **Table 1**). In running, $P_{\max}$ ($d = 0.131$, $p = 0.041$) and W ($d = 0.192$, $p = 0.030$) significantly increased from T1 to T2. The reliability of $t_{p\max}$ and $t_{p\max-3.5\%}$ attained rather 'poor' ICCs in cycling and running. The t_{100} demonstrated an 'excellent' ICC of 0.993 and significantly decreased between trials from 13.50 ± 1.23 s to 13.41 ± 1.26 s ($d = -0.07$, $p = 0.009$). Two-way ANOVA identified a significant time effect for $t_{p\max}$, $t_{p\max-3.5\%}$, and t_{lac} , whereas post-hoc differences were only significant in cycling. In cycling, $t_{p\max}$ ($d = -0.571$, $p = 0.024$) and $t_{p\max-3.5\%}$ ($d = -0.475$, $p = 0.025$) significantly decreased from T1 to T2 whereas t_{lac} demonstrated an increase ($d = 0.475$, $p = 0.025$). Hence, t_{lac} demonstrated 'moderate' reliability in cycling, whereas t_{lac} demonstrated 'excellent' reliability in running. Lactate kinetics interpolation parameters (A , k_1 and k_2) demonstrated different levels of reliability. For A , reliability was 'good' in cycling and 'moderate' in running. The increase constant k_1 was moderately reliable in cycling and poorly reliable in running. The removal constant k_2 demonstrated 'poor' reliability in both exercise modalities. In cycling, k_2 significantly decreased ($d = -0.767$, $p = 0.022$) whereas an increase in running was found ($d = 0.714$, $p = 0.023$). Hence, k_2 was the only parameter that demonstrated a significant time $\times$ modality interaction effect ($\eta_p^2 = 0.594$, $p < 0.001$). Whereas La_{pre} demonstrated 'moderate' reliability in cycling and 'poor' reliability in running, $La_{\max}$ and La_{Δ} demonstrated 'good' reliability in both exercise modalities. $\dot{V}La_{\max}$ demonstrated 'good' reliability in cycling and running. The mean difference between trials of $\dot{V}La_{\max}$ in cycling was $-0.003 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$ with limits of agreement ranging from -0.149 to $+0.143 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$ (► **Fig. 3a**). In running, the mean difference between trials was $-0.022 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$ with limits of agreement ranging from -0.181 to $+0.137 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$ (► **Fig. 3b**).

► **Table 1** Reliability, ANOVA and post-hoc comparisons for all parameters in cycling and running.

Parameter		T1	T2	ICC	Time		Modality		Time * Modality	
					η_p^2	P	η_p^2	P	η_p^2	P
m	Cycling	71.79 ± 7.01	71.90 ± 6.71	0.992	0.008	0.723	0.031	0.473	0.168	0.082
[kg]	Running	72.07 ± 6.73	71.83 ± 6.81	0.995						
P_{max}	Cycling	13.36 ± 2.15 ^b	13.34 ± 2.18 ^b	0.907	0.020	0.559	0.663 ***	< 0.001	0.035	0.442
[W · kg ⁻¹]	Running	11.62 ± 1.29 ^{a,b}	11.79 ± 1.31 ^{a,b}	0.965						
W	Cycling	164.96 ± 25.08 ^b	164.24 ± 26.25 ^b	0.899	0.001	0.889	0.746 ***	< 0.001	0.023	0.537
[J · kg ⁻¹]	Running	128.89 ± 5.42 ^{a,b}	130.00 ± 6.11 ^{a,b}	0.927						
t_{pmax}	Cycling	2.99 ± 0.54 ^{a,b}	2.69 ± 0.51 ^{a,b}	0.454	0.374 **	0.005	0.547 ***	< 0.001	0.067	0.286
[s]	Running	2.31 ± 0.30 ^b	2.19 ± 0.32 ^b	0.213						
$t_{pmax-3.5\%}$	Cycling	4.82 ± 0.83 ^{a,b}	4.44 ± 0.77 ^{a,b}	0.592	0.335 **	0.009	0.657 ***	< 0.001	0.051	0.355
[s]	Running	3.53 ± 0.46 ^b	3.37 ± 0.54 ^b	0.242						
t_{lac}	Cycling	10.18 ± 0.83 ^a	10.56 ± 0.77 ^a	0.592	0.250 *	0.029	0.045	0.386	0.082	0.233
[s]	Running	9.97 ± 1.46	10.04 ± 1.60	0.904						
A	Cycling	7.57 ± 1.97	7.44 ± 1.89	0.839	0.060	0.312	0.017	0.594	0.117	0.152
[mmol · l ⁻¹]	Running	7.44 ± 1.07	7.95 ± 1.66	0.567						
k_1	Cycling	0.77 ± 0.13 ^b	0.75 ± 0.14	0.550	0.032	0.466	0.275 *	0.021	0.012	0.649
[min ⁻¹]	Running	0.95 ± 0.20 ^b	0.89 ± 0.38	0.346						
k_2	Cycling	0.042 ± 0.014 ^{a,b}	0.032 ± 0.012 ^a	0.171	< 0.001	0.962	0.586 ***	< 0.001	0.594 ***	< 0.001
[min ⁻¹]	Running	0.016 ± 0.014 ^{a,b}	0.026 ± 0.014 ^a	0.204						
La_{pre}	Cycling	1.06 ± 0.55 ^b	0.89 ± 0.24 ^b	0.573	0.100	0.188	0.566 ***	< 0.001	0.024	0.530
[mmol · l ⁻¹]	Running	1.59 ± 0.55 ^b	1.54 ± 0.48 ^b	0.335						
La_{max}	Cycling	7.16 ± 1.56 ^b	7.21 ± 1.54 ^b	0.870	0.023	0.532	0.544 ***	< 0.001	0.015	0.177
[mmol · l ⁻¹]	Running	8.70 ± 1.43 ^b	8.44 ± 1.30 ^b	0.808						
La_{Δ}	Cycling	6.10 ± 1.63 ^b	6.31 ± 1.56	0.867	< 0.001	0.974	0.247 *	0.031	0.140	0.114
[mmol · l ⁻¹]	Running	7.10 ± 1.09 ^b	6.90 ± 1.08	0.783						
$\dot{V}La_{max}$	Cycling	0.60 ± 0.16 ^b	0.60 ± 0.15 ^b	0.894	0.057	0.325	0.383 **	0.005	0.028	0.493
[mmol · l ⁻¹ · s ⁻¹]	Running	0.73 ± 0.16 ^b	0.71 ± 0.16 ^b	0.868						

Values are expressed as mean ($\bar{x}$) and standard deviation (SD). * significant factor ($p \leq 0.05$); ** significant factor ($p \leq 0.01$); *** significant factor ($p \leq 0.001$); ^a significant difference between T1 and T2 ($p \leq 0.05$); ^b significant difference between cycling and running ($p \leq 0.05$); post-hoc comparisons based on Bonferroni's adjustment. A = amplitude parameter describing post-exercise lactate kinetics of the all-out exercise test; k_1 = velocity constant describing the exchange of lactate from the previously active muscles; k_2 = velocity constant describing the removal of lactate during passive recovery; La_{pre} = lactate concentration immediately before the all-out exercise test; La_{max} = maximal lactate concentration after the all-out exercise test; La_{Δ} = increase in lactate concentration after all-out exercise; La_{pre} = lactate mass; P_{max} = peak power output during all-out exercise tests; t_{lac} = lactic time interval; $t_{pmax-3.5\%}$ = time to attain peak power output; $t_{pmax-3.5\%}$ = time when peak power output decreased by 3.5% [used as the time at the beginning of exercise for which no lactate formation is assumed (t_{lac})]; $\dot{V}La_{max}$ = maximal lactate accumulation rate (lactic power); W = work attained during all-out exercise tests.

Differences between modalities

Exercise modality had a significant effect on $P_{\max}$, W , $t_{p\max}$, $t_{p\max-3.5\%}$, k_1 , k_2 , La_{pre} , $La_{\max}$, La_{Δ} and $\dot{V}La_{\max}$ (► **Table 1**). Even though there was a significant difference between exercise modalities found in k_1 , k_2 and La_{Δ} at T1, there was no significant difference after familiarization. At T2, La_{pre} ($d = 1.713$, $p < 0.001$), $La_{\max}$ ($d = 0.863$, $p = 0.001$) and $\dot{V}La_{\max}$ ($d = 0.709$, $p = 0.016$) were significantly higher in running whereas $P_{\max}$ ($d = -0.862$, $p < 0.001$), W ($d = -1.797$, $p < 0.001$), $t_{p\max}$ ($d = -1.174$, $p = 0.006$) and $t_{p\max-3.5\%}$ ($d = -1.609$, $p < 0.001$) were significantly lower in running compared to cycling (► **Fig. 2**). Whereas the mean difference between exercise modalities in $\dot{V}La_{\max}$ was around $0.1 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$, limits of agreement ranged from -0.224 to $0.437 \text{ mmol} \cdot \text{l}^{-1} \cdot \text{s}^{-1}$ (► **Fig. 3c-d**). However, the differences in $\dot{V}La_{\max}$ between cycling and running demonstrated a 'good' reproducibility between trials ($\text{ICC} = 0.754$). There was no significant difference in t_{lac} between cycling and running at T1 ($d = -0.177$, $p = 0.620$) and T2 ($d = -0.414$, $p = 0.256$).

Correlation analyses

There was a significant correlation between $P_{\max}$ and $\dot{V}La_{\max}$ in both cycling ($r = 0.719$, $p = 0.001$) and running ($r = 0.735$, $p = 0.001$). The same applies for W , which correlated with $\dot{V}La_{\max}$ in cycling ($r = 0.756$, $p < 0.001$) and running ($r = 0.717$, $p = 0.001$) as well. La_{Δ} correlated with $P_{\max}$ in cycling ($r = 0.747$, $p < 0.001$), but not in running ($r = 0.290$, $p = 0.243$). As with $P_{\max}$, W and La_{Δ} did correlate in

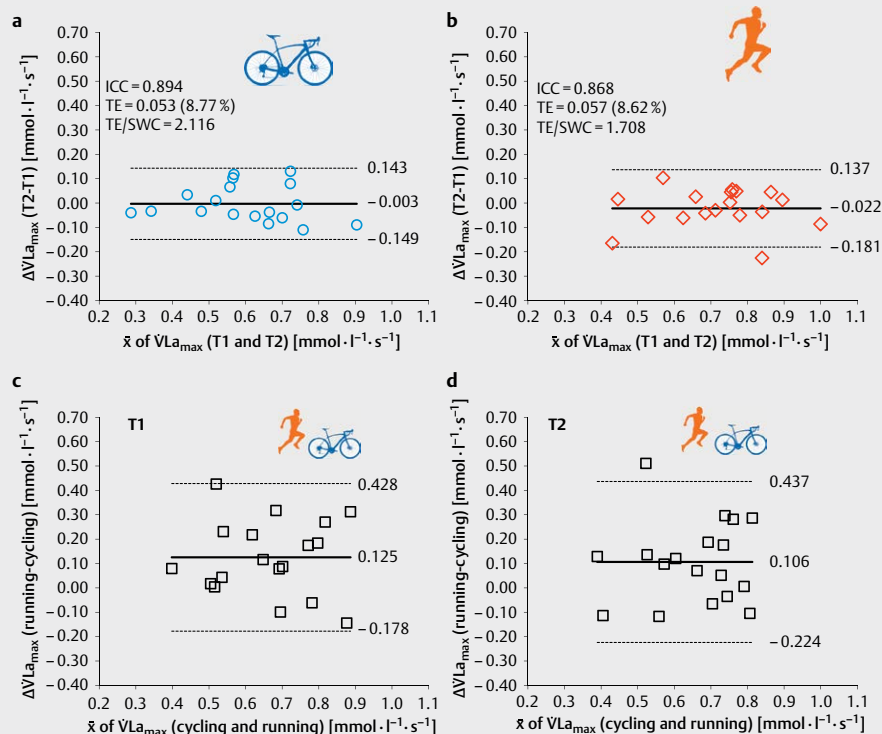
cycling ($r = 0.741$, $p = 0.001$), but not in running ($r = 0.251$, $p = 0.315$). In running, t_{100} correlated with $P_{\max}$ ($r = -0.969$, $p < 0.001$), W ($r = -0.898$, $p < 0.001$) and $\dot{V}La_{\max}$ ($r = -0.775$, $p < 0.001$). There was a significant correlation between cycling and running in $P_{\max}$ ($r = 0.811$, $p < 0.001$) and W ($r = 0.659$, $p = 0.003$). However, La_{Δ} ($r = 0.432$, $p = 0.073$) and $\dot{V}La_{\max}$ ($r = 0.418$, $p = 0.084$) did not correlate between exercise modalities.

Discussion

The aims of this study were to compare $\dot{V}La_{\max}$ and $P_{\max}$ between cycling and running in terms of reliability, differences between and correlations among exercise modalities.

Reliability and differences between trials

The reliability attained in cycling and running was in accordance with previous research [7, 23, 28]. $\dot{V}La_{\max}$ and $P_{\max}$ attained 'good' to 'excellent' reliability in both cycling and running. For the cycling data, the results of the reliability analysis have already been reported in a previous article [23], which are similar to the findings of Adam et al. (2015), who used another type of ergometer (Lode Excalibur Sport, Lode, Groningen, Netherlands) [28]. Rylands et al. (2016) highlighted that the cadence applied in a sprint test in cycling affects the power-time course and thus $P_{\max}$ as well as $t_{p\max}$



► **Fig. 3** Bland-Altman plots of $\dot{V}La_{\max}$ differences between trials (a-b) and modalities (c-d). **a** Differences of $\dot{V}La_{\max}$ between T1 and T2 in cycling **b** Differences of $\dot{V}La_{\max}$ between T1 and T2 in running **c** Differences of $\dot{V}La_{\max}$ between cycling and running at T1 **d** Differences of $\dot{V}La_{\max}$ between cycling and running at T2. Mean differences (solid lines) and limits of agreement (± 1.96 SD, discontinuous lines). ICC = intra-class correlation coefficient; SWC = smallest worthwhile change [based on a small Cohen's d effect size ($d = 0.20$)]; TE = typical error; TE/SWC = ratio of TE compared to SWC.

[30]. Despite the fact that $P_{\max}$ attained 'excellent' reliability in cycling, one should be aware of the influence of the test conditions.

In running, reliability was slightly lower when compared to the preliminary study [7]. This might be due to the fact that the time between measurements was two days in the preliminary study and seven days in this study. From a practical perspective this indicates that exercise tests should be performed as close as possible to the date of interest, even though a delay of one week might not be harmful for data interpretation.

Lower reliability of the lactate response in running might be due to the fact that standardization was easier to apply in cycling. In running, the participants decreased their t_{100} predominantly by familiarization to the start-maneuver [7]. This assumption is justified by the facts that $P_{\max}$ was significantly higher at T2 and $t_{p\max}$ tended to decrease in the same amount as t_{100} (around -0.1 s), even though differences between trials were not statistically significant for $t_{p\max}$. However, a preliminary study indicated that the velocity and power at the very end of the sprint significantly increased after initial familiarization in a similar sample of endurance athletes [7]. Given the data provided here, we can only speculate whether the decrease in t_{100} was due to an improved start and/or finish. However, given the high reliability and very small effect-size, the improvement in t_{100} seems to be negligible.

A decreased $t_{p\max}$ and consequent increase in t_{lac} after familiarization in cycling might be explained by an improved recruitment and coordination of fast-twitch muscle fibers. This familiarization effect was found to be lower in running, which might be due to a higher muscular strength required at the very start in cycling compared to running [30, 31]. Lactate kinetics interpolation parameters demonstrated rather low reliability in both modalities, which was most severe in lactate removal constant (k_2). Since k_2 is also influenced by various tissues [32], the decrease in cycling and increase in running should not be over-interpreted. To improve the reliability of k_2 and other interpolation parameters, an increase of the post-exercise period (e. g. up to 20 or 30 min) would provide a higher number of blood samples required for a more detailed description of lactate kinetics as highlighted elsewhere [33].

Differences between modalities

Differences between cycling and running might be due to sport-specific muscle recruitment, testing conditions and the participants' training background in the particular modality. Whereas the knee flexors and hip extensors are not fully activated during all-out cycling, these muscles are highly activated in all-out running [34, 35]. Due to a higher working muscle mass and lower lactate distribution volume, lactate response might be higher in running compared to cycling [36, 37]. Similarly, findings of a previous study demonstrated that $\dot{V}La_{\max}$ is higher in cycling when compared to handcycling indicating that this parameter is extremity-specific [23]. A higher La_{pre} in running might also be due to the warm-up routine [7]. Running drills and strides were performed in order to prepare participants for all-out sprints and prevent injuries, but might lead to a net lactate accumulation prior to the sprint tests. Lactate kinetics interpolation parameters demonstrated no difference between modalities after familiarization, which might be due to their high variation and rather low reliability. The lower $P_{\max}$ and

W observed in running compared to cycling can be explained by the calculation approach based on the athletes' velocity profiles [7]. In running, the athletes have to move their limbs around the center of mass in order to increase and maintain velocity. Moreover, additional (internal) work is neglected by using horizontal velocity [38]. Nonetheless, previous research performing all-out running on a non-motorized treadmill demonstrated a similar difference in $P_{\max}$ between running and cycling [39]. In cycling, a high initial force has to be applied on the pedals, which might delay an increase in pedal frequency and cause a longer $t_{p\max}$ in cycling compared to running [30]. $\dot{V}La_{\max}$ was found to be higher in running (+ 25 %) with a large standard deviation (± 47 %) and poor correlation ($R^2 = 17.4$ %) between modalities. This is in contrast to studies assuming a higher net rate of lactate formation in cycling compared to running [40] and rowing exercise [41]. One might argue that the higher $\dot{V}La_{\max}$ in running is influenced by the (slightly) lower exercise time since La_{Δ} (numerator) was not significantly different at T2. However, since t_{lac} (denominator) was not significantly different either, it is unlikely that this is the only influence on $\dot{V}La_{\max}$ differences. Thus, differences between modalities seem to be influenced by both, the numerator and the denominator of $\dot{V}La_{\max}$ calculation. Since these differences demonstrated good reproducibility, it seems that the glycolytic metabolism adapts rather modality-specific to training stimuli accounting for the intra-individual difference in $\dot{V}La_{\max}$ between cycling and running. Future intervention studies need to verify this hypothesis for $\dot{V}La_{\max}$ analogous to findings in $\dot{V}O_{2\max}$ and lactate threshold. Furthermore, knowing that net lactate formation tends to be higher in running (compared to cycling) is highly important for mathematical simulation approaches trying to implement sport-specific estimations of maximal lactate steady-state by means of $\dot{V}O_{2\max}$ and $\dot{V}La_{\max}$.

Correlation analyses

Sprint performance and lactic power demonstrated a close relationship in both modalities indicating the validity of $\dot{V}La_{\max}$ in cycling and running. Since $P_{\max}$ correlated between modalities and $\dot{V}La_{\max}$ and La_{Δ} did not, it can be assumed that the phosphagenous energy system demonstrates a higher transfer between cycling and running and that the glycolytic metabolism requires rather modality-specific adaptations [37, 42]. If so, this could be an important finding for the training and testing of multisport athletes.

Conclusions

$\dot{V}La_{\max}$ is highly reliable in both exercise modalities and higher in running compared to cycling. However, the large inter-individual difference and low correlation between cycling and running indicates that $\dot{V}La_{\max}$ is sensitive to exercise modality and seems to depend on training experience and particular capabilities. Hence, $\dot{V}La_{\max}$ should be determined sport-specifically. The findings of this study can be used to assess the effect of deliberate training (high volume or high intensity) on $\dot{V}La_{\max}$ in cycling and running. Furthermore, these findings support mathematical simulation approaches attempting to estimate maximal lactate steady-state in running by means of $\dot{V}O_{2\max}$ and $\dot{V}La_{\max}$.

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Conflict of interest

The authors declare that there is no conflict of interest regarding this article.

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